

**Plant Animal Interactions and Nesting Behavior of Dusky-footed
Woodrats (*Neotoma fuscipes*) in Relation to Vegetation and
Environmental Variables in the Bay Area**

Introduction

Dusky-footed woodrats (*Neotoma fuscipes*) are native to California and Oregon and can be found throughout the Bay Area. These New World rodents play a significant role in their ecosystems, both as prey (supporting predators like owls, hawks, *Lynx rufus* (bobcats), and *Canis latrans* (coyotes)) and as builders and foragers that influence plant dynamics. They are avid generalist feeders of all sorts of plant material, like fungi, bark, seeds, and plants. Their nesting behavior is incredibly fascinating; they build nests made out of stick/wood material that can get up to ‘5 feet in height and eight feet in diameter’ (NWF). They also display specific interactions with the flora in their habitat, specifically with bay laurel trees (*Umbellularia californica*), by lining their nests with its leaves. It is thought that these leaves could potentially have some sort of fumigating properties that ward off mites and bacteria and ticks.

Umbellularia californica itself has an infamous role in ecosystems, especially around the Bay Area. This tree is a vector species for the pathogen *Phytophthora ramorum*, also known for causing Sudden Oak Death (SOD). SOD is a disease that primarily affects white oaks and tanoaks, and causes them to die (longer than the name suggests - sometimes upwards of two years). *Umbellularia californica* isn’t impacted by carrying this pathogen, but it easily spreads through close proximity to susceptible species. UC Berkeley’s Forest Pathology and Mycology Lab states that the pathogen is “estimated to have been introduced into California 20-25 years ago from unknown regions of the world” (Matteo Lab). Since then it has killed roughly around one million tree species in California, although the exact number is unknown (CISR).

On all field classes where I saw *Neotoma fuscipes* nest signage, it became pretty clear that there was a close relationship between the woodrat habitat and oak woodland habitat. Noticing this while also learning about their interactions with *Umbellularia californica* made me wonder how this impacted dusky-footed woodrats and their habitats.

Research Questions & Hypothesis

Question: What environmental and botanical factors influence where dusky-footed woodrats build their nests?

- Are dusky-footed woodrats aware of diseased *Umbellularia californica* trees, and do they alter their nesting habitats because of it?

Hypothesis: Dusky-footed woodrats preferentially build nests near specific tree species (*Quercus agrifolia* or *Umbellularia californica*), and factors like sun exposure, elevation, and disease presence (sudden oak death) also affect nest-site selection and behavior changes in respect to the collection of *Umbellularia californica*.

Methods and Materials

A. Study Sites

The two study sites I chose to do my observations at were Tilden Regional Park and Briones Regional Park. For Briones Regional park, there were two locations I looked at:

37.94562, - 122.13412

37.92807, - 122.15429

For Tilden Regional Park, I also made observations in two different locations:

37.90348, - 122.25917

37.89175, - 122.24092

Based on the chart from the lab website: [https://nature.berkeley.edu/matteolab/?](https://nature.berkeley.edu/matteolab/?page_id=7878)

[page_id=7878](https://nature.berkeley.edu/matteolab/?page_id=7878), I knew that there had been previous tests in the East Bay, and SOD had been found.

SITES	DATES	# of NESTS OBSERVED
Briones 2	4/19/25	5
Tilden 1	4/21/25	6
Briones 1	4/22/25	0
Tilden 2	4/26/25	10
Briones 2	5/4/25	11

Data Collection

- Survey of 32 individual woodrat nests.
 - For each woodrat nest identified, I aimed to spend at least three minutes observing the area, surrounding vegetation, perimeter inspection of bay laurel leaves, inspection of nearby trees for signs of SOD, and to record my observations.
- Latitude and longitude mapping of nests for spatial analysis.
- Recorded variables:
 - tree species near each nest
 - Elevation and slope.
 - Sun/light exposure (0%, 50%, 100%).
 - I decided to record this variable in these three fixed increments because the amount of exposure a certain nest received is subjective and hard to measure, and generally speaking, the nest was either completely out in the open with no canopy, completely shaded, or some mix of both.
 - Whether or not a nest was built near diseased *U. californica* and/or *Q. agrifolia*.
 - I recorded these variables by physically measuring distances between nests and *U. californica* and *Q. agrifolia* trees, and taking note if there weren't any species within ten yards of the nest.
 - Whether there were signs of *U. californica* leaves surrounding nests
 - I collected this data by getting as close as possible to each nest, walking around the perimeter, and seeing if I could find any chewed up or whole bay laurel leaves.
 -

- Nest incline on the landscape (0 degrees = flat, 65% = cliffside).
 - Like the exposure variable, I created set data points for this variable. If the nest was on flat ground, the incline was indeed 0 degrees. However, if it was on an obvious cliff, I decided to set the incline to be 65 degrees straight. Anything in the middle of those two options I used my best discernment.

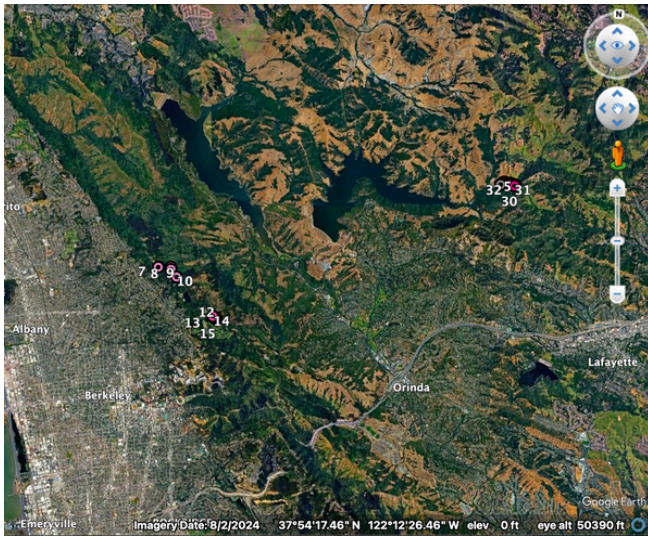
Materials

- GPSCoordinates mobile app
 - I used this app to screenshot the latitude and longitude coordinates whenever I reached a new nest area.
- Google Sheets
 - I used google sheets to sort and organize my data, and create line graphs and bar charts to visualize my data.
- Google Earth Pro
 - I used Google Earth Pro to view data points spatially, as well as measure out the distance between points and elevation changes.
- Seek
 - I have a pretty good understanding of how to identify native woody plants, especially California bay laurel, coast live oak, and Poison Oak (*Toxicodendron diversilobum*), but I did use Seek to identify a plant speices I was unsure about.
- Tape Measure
 - I used a tape measure when appropriate to measure bay laurel/cost live oak trees close enough to a nest
- UC Berkeley's Forest Pathology and Mycology Lab
 - I used resources from this lab website to learn about SOD locally, how to identify leaves, and more.

Data Analysis and Results

Spatial Visualization of 32 *Neotoma fuscipes* Nesting Sites

Figure 1: Coordinate Mapping



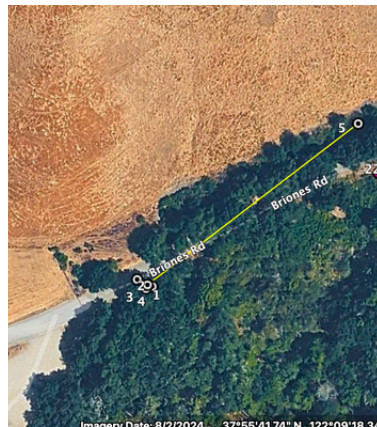
A



B



C



D

Figure 1. A: This image depicts the spatialization between all 32 data points. These points span across the East Bay. B: Image B depicts the data points 1 - 5, and 22 - 32.



E

Figure 1 (cont.) As calculated by Google Earth Pro using the 3D path tool, the total distance between nests 1-5 is 366.19 feet. The total distance between nests 22 - 32 is 1276.81 feet. This is in Briones Regional Park C: Image C shows the nests 6 - 11. The distance between the points, in total, is 3126.17 feet.

C is showing Tilden Regional Park. D: This image is a close up of data points 1 - 5 (Briones Regional Park). E: Finally, E shows the data points 12 - 21. The total distance between these nests is 1704.92 feet. (Tilden Regional Park)

Environmental Preferences with Nesting Strategies

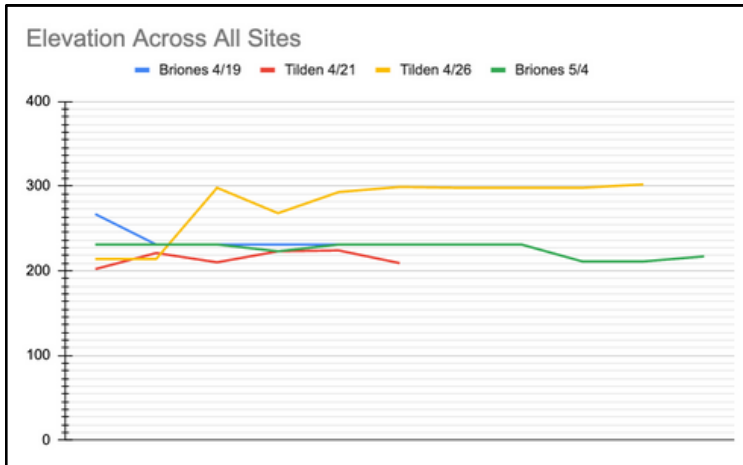


Figure 2: General Elevation

Figure 2: General Elevation. This graph depicts the general trends in elevation of the surveyed nest sites, with color coding tags for each location.

Figure 3: Comparison of Elevation, Exposure, and Incline. This stacked bar chart displays three variables at once in respect to each numbered nest: elevation (in meters), sun/light exposure (in percent), and incline (degrees).

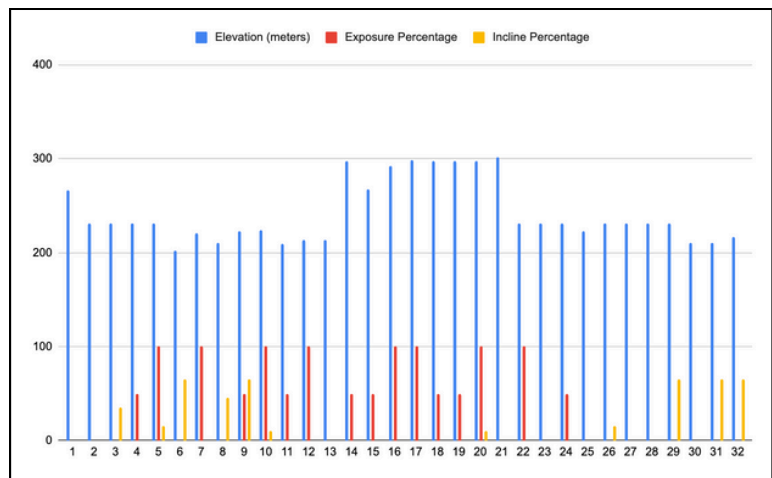


Figure 3: Comparison of Elevation, Exposure, and Incline

The data from figure three suggests that there is little to no correlation between elevation, sun/light exposure, and degree of incline. There is a slight trend where nests 14, 15, 16, 17, 18, 19, and 20 have a noticeably higher elevation than the rest of the data set, and these points all have 0 degrees of incline (except for nest 20 with minimal inclination). However, when I created a T-Test for this data, I get the results: T-statistic: -0.669 and P-value: 0.508, meaning there is a very weak negative relationship, and that there is no statistically significant correlation.

Tree Type Associations Data

Figure 4: This line graph combines the graphing of amount of *T. diversilobum* (on a scale between 0 - 10) within a five meter radius surrounding each nest site, with data points displaying what trees/ group of trees are also nearby the nests.

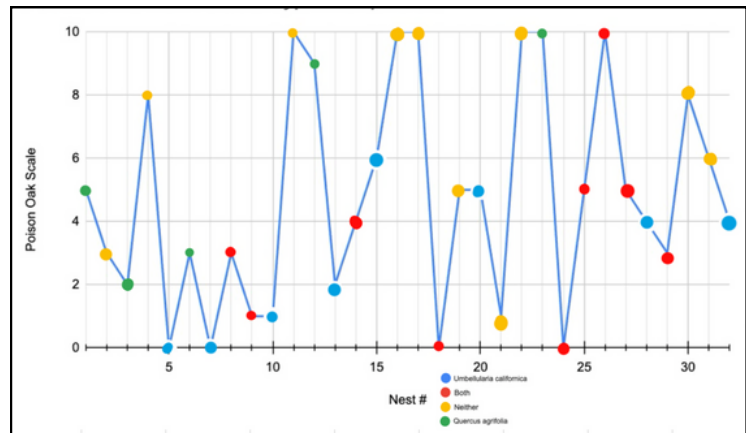


Figure 4: *T. diversilobum* Rating vs. Tree Type

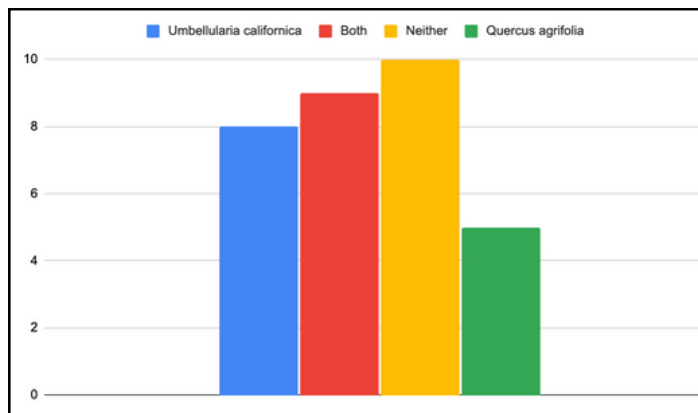


Figure 5: This histogram displays how many nests were directly under an *U. californica*, a *Q. agrifolia*, both, or neither.

Figure 5: Nest Proximity to *U. californica* and Proximity to *Q. agrifolia*

These results show that the category with the single most nests is neither. This at first surprised, until I looked at the data more comprehensively. This does not mean that nests are more likely to be built under a totally different tree or a new tree all together, but instead it is more likely for a nest to be built under/within at least one of the two trees, if not both. Figure 4 results show a seemingly random spread of information. The only trend that I am able to extract from this graph is a slight correlation between *U. californica* and low amounts of *T. diversilobum*. All *T. diversilobum* levels are 6 and below for nests built next to only *U. californica*. The t-statistic = -2.48 and the p-value = 0.023. This suggests that the difference is statistically significant; nests next to *U. californica* have lower poison oak presence compared to other nests.

Sudden Oak Death Observation Data

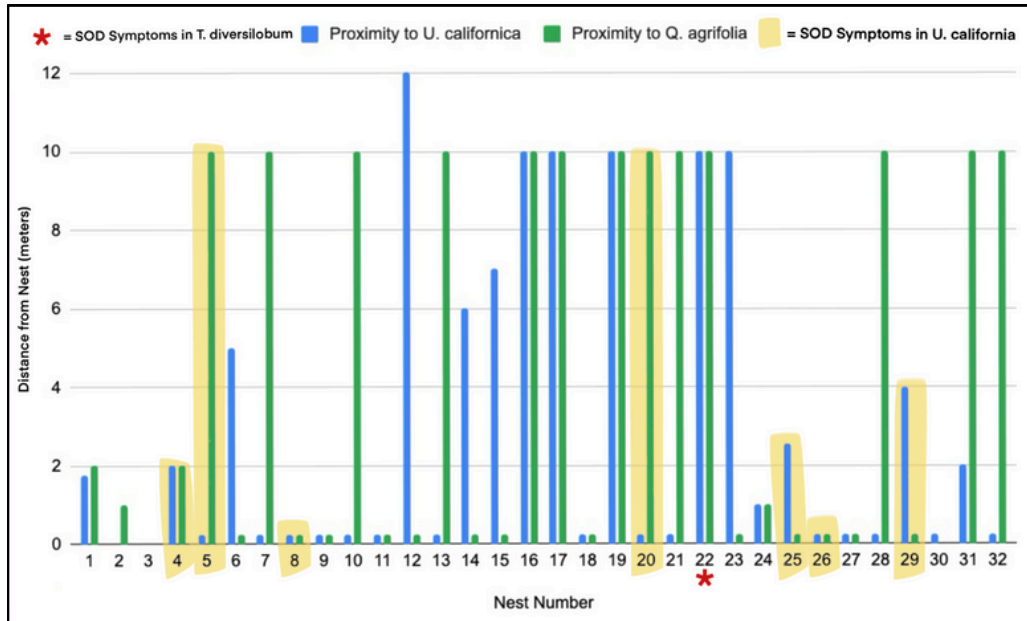


Figure 6: SOD Presence by Nest Proximity to *U. californica* vs. *Q. agrifolia*

Figure 6 is a stacked bar chart that depicts multiple variables and how they interact with each other. To start, the blue and green lines represent nest proximity to *U. californica* and *Q. agrifolia* respectively. The height of each bar is the distance away from the nest; if the bar reaches 10 meters, they were either 10 or more meters away from a certain nest and therefore are not relevant to the results. If proximity was with half a meter (nest is on top of, through, within root structure of tree), I set the proximity to be .25 meters to show on the graph. If highlighted in yellow, *U. californica* showed signs of SOD.

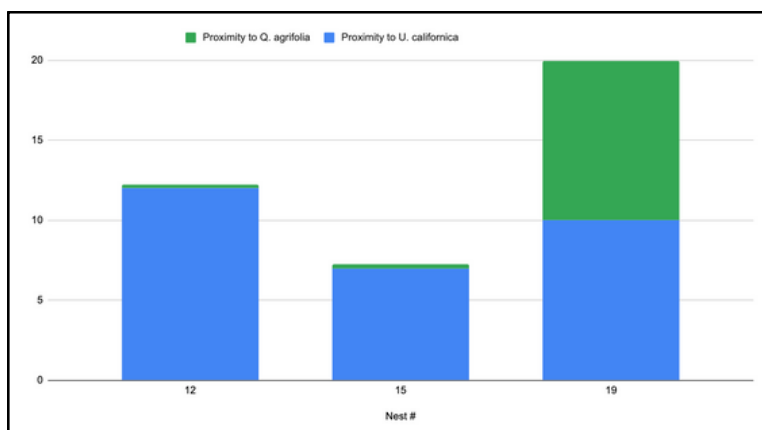


Figure 7: Tree Proximity and *U. californica* Leaf Presence at Nests

Figure 7 is a stacked bar chart showing the relationship between nests that had visible signs of *U. californica* around the perimeter of the nest, and the nest's proximity to *U. californica* and *Q. agrifolia*.

The results from Figure 6 and Figure 7 ultimately show little correlation between nests near *U. californica* infected with SOD and nests near *U. californica* not infected with SOD. I would like to note that each nest near a bay laurel tree that *does* appear to be infected with SOD is relatively close in proximity to the nest. However, there are many instances where a nest is similarly close in proximity to a bay laurel tree that is not infected with SOD. The red asterisk is to point out that I did initially think that I identified SOD signage in the branches of Nest 22 because it was incredibly red, but I misidentified.



In Figure 7, I strictly compared nests where I found bay laurel leaves around the nest (unless the nest was directly underneath a bay laurel tree to prevent mistaking fallen leaves as purposefully placed) with proximity to both bay laurel and coast live oak. The chart shows that at all three nest sites, bay laurel is at least 7 meters away from the nest, with coast live oak being much, much closer. This was fascinating because this proved that the bay laurel leaves I found strewn around the nests most likely were intentionally brought.



The bay laurel leaves to the left were found at Nest 19. It looked like an entire branch was dragged over and carefully threaded through sticks and pieces of wood. They appeared to have been there for a while because they are crunchy and brown, like they've been dead for a while.

The bay laurel leaves to the left were found at Nest 15. These similarly look like they had been around the nest for a long time. These were found separately, not connected on a branch like those found at Nest 19. They were also bigger and felt slightly further along on their decomposing journey.



U. californica Leaf Analysis



Non-infected Leaves at Nest 28



SOD Non-infected at Nest 16



Undetermined Leaves at Nest 18



SOD Infected at Nest 20



SOD Infected at Nest 4



SOD Infected at Nest 5

Discussion

The aim of this research project was to explore what environmental and botanical factors influence nest-site selection by *Neotoma fuscipes*, the dusky-footed woodrat. Specifically, this research examined whether tree species, environmental variables (elevation, sun exposure, incline), disease presence (SOD), and other foliage (poison oak) played a role in nest placement/bay laurel usage.

Contrary to the original hypothesis, the data showed no statistically significant relationship between elevation, sunlight, or incline and nest location. While I was able to identify a couple trends just by the look of the graphs, like grouping of higher elevation nests with low incline, the t-tests ultimately confirmed that these variables had no meaningful influence on woodrat nest placement. When assessing tree association, the single most frequent category I found *Neotoma fuscipes* nests was under ‘neither’ bay laurel nor coast live oak. However, most nests were still built within close proximity to at least one of these species, and many under both. This shows that perhaps these woodrats have a more flexible tree association rather than strong exclusivity. The strongest statistical finding came from examining the relationship between *U. californica* and *T. diversilobum*. Nests near only *U. californica* showed significantly lower poison oak levels, and this suggests that bay laurel proximity may coincide with reduced exposure to toxic understory plants like poison oak. Why exactly this occurs is a research question I would be interested in exploring further. This was also a result that I had not anticipated or even considered, and it was only by curiosity that I decided to compare and contrast understory vegetation near woodrat nests with the two focus tree species. An analysis of sudden oak death infection in *U. californica* revealed no clear preference or avoidance pattern in nest placement between infected and uninfected trees. This result suggests that woodrats either cannot detect SOD or do not prioritize it in nest-site decision-making. It's also possible that if a bay laurel is infected with SOD, it does not dramatically alter the proposed chemical/fragrant qualities that are attractive to woodrats and making their nests.

One of the most intriguing findings for me came from direct observations of bay laurel leaves intentionally brought to nests. *U. californica* leaves were found within the structure of nests that were not near bay laurels, but rather near *Q. agrifolia*. This implies intentional gathering behavior, possibly for the chemical, structural, or anti-parasitic properties mentioned earlier. That the leaves were brittle and partially decomposed suggests they had been there for some time, which strengthens the idea that these placements were deliberate and not the result of incidental leaf fall. Of course, this is a known behavior, but it was still fascinating to be able to observe it firsthand.

Limitations

There are several potential limitations that I became aware of throughout this research project, and that should be acknowledged when interpreting the results of this study:

Sample Size: The study was limited to a sample of 32 nests, which may not fully represent the variability in dusky-footed woodrat nesting behavior across the Bay Area and especially in the greater California.

Seasonal Bias: Data collection occurred during a single season, and a relatively short window within that season, which may lead to some seasonal shifts in woodrat vegetation/nest patterns not being reflected. For example, the presence or visibility of certain foliage, like poison oak or deciduous trees, varies throughout the year and may influence both nest visibility and habitat preferences.

Observational Error: There is always potential for human error in field identification of tree species, and even identifying nests themselves. Additionally, evaluating the presence of sudden oak death based solely on visual indicators may not always be accurate without lab testing.

Woodrat Activity Uncertainty: The presence of a nest does not guarantee that there were woodrats actively residing in them. Some nests may have been abandoned. Without confirmation of active use through motion sensors or direct observation, it is difficult to determine whether environmental factors influenced things nest placement before or after the nest was built. For example, a tree could become infected with SOD after a nest has

been constructed, and this was extremely difficult to discern strictly through observation. Partial Nest Data: While bay laurel leaves were occasionally observed around nests, I was not able to collect data inside the nests, which is where woodrats often put the leaves. This presents a limitation in understanding the full scope of materials intentionally gathered or used by the woodrats.

Sources

“Dusky-Footed Woodrat.” National Wildlife Federation, www.nwf.org/Educational-Resources/Wildlife-Guide/Mammals/Dusky-Footed-Woodrat#:~:text=The%20abandoned%20homes%20of%20dusky,%2C%20greens%2C%20and%20inner%20bark. Accessed 7 May 2025.

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“Sudden Oak Death.” Center for Invasive Species Research, 28 Sept. 2022, civr.ucr.edu/invasive-species/sudden-oak-death#:~:text=It%20is%20estimated%20that%20the,of%20central%20and%20northern%20California.